AKKA

Scale AI, Efficiently
The Agentic Systems Platform

Akka vs. Microsoft Foundry (formerly Azure AI Foundry)

A comparison for teams building agentic AI

October 2026 · Microsoft has renamed the product Microsoft Foundry; this card covers the Foundry Agent Service.

Foundry is a collection of Azure services to integrate, not a platform — and Foundry provides no automatic failover or disaster recovery for its Agent Service. Microsoft's own documentation states that Foundry has no automatic failover, that the Agent Service does not replicate state, and that the recovery point for agent state "can be total loss." The Akka Agentic AI Platform delivers agents, memory, streaming, APIs, and governance as one runtime that guarantees resilience and scalability, at a contractual 99.9999% SLA, sub-1-minute RTO, and zero-byte RPO.

No failover	99.9999%	5+	Total loss
FOUNDRY AUTOMATIC FAILOVER OR DR	AKKA SLA, WITH INDEMNITIES	AZURE SERVICES TO STITCH TOGETHER	DOCUMENTED STATE RECOVERY POINT

DIMENSION	FOUNDRY	AKKA
What it is	An agent service orchestrating a set of separately provisioned Azure services	The Akka Agentic AI Platform — a platform that guarantees resilience and scalability, on one runtime
Scope	Agent runtime + bring-your-own Cosmos DB, AI Search, Storage, Key Vault, OpenAI, API Management, Monitor — integrated and operated by you	Agents, memory, streaming, APIs, orchestration, observability, and governance on one runtime
Availability SLA	None stated in Microsoft's Foundry HA/DR documentation; no automatic failover	99.9999% — entire platform, backed by indemnities
HA / DR	No automatic failover; no built-in DR; no active-active multi-region replication	Active-active HA/DR; sub-1-min RTO; zero-byte RPO
Recovery point	"Can be total loss"; cross-region state lost on failback	Zero-byte RPO; state fully preserved
Governance / EU AI Act	Guardrails at agent input, tool call, tool response, and output (preview) for Foundry Agent Service agents; Control Plane stop and block; Purview templates for EU AI Act, ISO/IEC 42001, ISO/IEC 23894, NIST AI RMF	Aspect-woven runtime enforcement + full pre-production governance
Portability	Azure identity, gateway, and data services; moving off means rebuilding	Any cloud, on-prem, or Akka cloud; portable specs; sovereign cloud
Cost model	Per-token / PTU OpenAI + per-RU Cosmos + per-tier AI Search + gateway + per-GB logging	Shared compute; up to 90% lower infrastructure; fixed annual fee
Certifications	Inherited Azure certifications + Purview Compliance Manager templates	19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF)

A Collection of Services to Integrate, Not a Platform

A production agent on Foundry is assembled from multiple, separately provisioned, separately billed Azure services. In the Agent Service Standard deployment mode, Microsoft's own setup documentation requires the customer to bring and operate an Azure Cosmos DB account (minimum 3,000 RU/s), an Azure AI Search resource, an Azure Storage account, and an Azure Key Vault — and a

realistic production agent also needs Azure OpenAI, an API Management gateway, and Azure Monitor / Application Insights. (Standard agent setup)

Azure service	Role in an agent	What the customer owns
Azure OpenAI	Model inference	Per-token billing or PTU capacity reservation
Azure Cosmos DB	Agent state / thread memory	Per-RU billing; ≥3,000 RU/s; backup, replication, failover
Azure AI Search	Vector store / retrieval	Per-tier billing; index rebuild on recovery
Azure Storage	Uploaded files, attachments	Redundancy and failover configuration
Azure API Management	Generative AI gateway, load balancing, circuit breaker	Multi-region gateway deployment and policy
Azure Monitor / App Insights	Observability	Per-GB ingestion; multi-region instances

Microsoft's documentation is explicit that the customer operates durability: "Microsoft and you jointly operate the Foundry Agent Service… You own the durability of stateful dependencies (Azure Cosmos DB, Azure AI Search, Azure Storage)." (HA/DR for Foundry) The integration, the reliability, and the recovery across these seams are the customer's responsibility. Akka delivers agents, durable memory (4ms reads / sub-10ms writes), real-time streaming, an API layer, orchestration, observability, and governance on one runtime — provisioned once, operated as one system, under one SLA.

Foundry Provides No Automatic Failover or Disaster Recovery

Microsoft's current documentation states this directly (HA/DR page updated September 2026; Agent Service DR page updated May 2026):

- "Foundry itself doesn't provide automatic failover or disaster recovery." (HA/DR for Foundry)
- "Agent Service doesn't provide built-in disaster recovery capabilities. It doesn't replicate state, create backups, or support point-in-time restore… no supported method for active-active, multi-region replication." (Agent Service DR)
- "The recovery point for stateful content can be total loss."

Metric	Foundry Agent Service	Akka
Availability SLA	None stated in Microsoft's Foundry HA/DR documentation	99.9999%
HA/DR mode	Warm-standby reconstruction; no automatic failover	Active-active, automatic
RTO	"30 or more minutes per project"	Sub-1 minute
RPO	"Can be total loss"	Zero byte
Multi-region	"No supported method for active-active"	Active-active across regions
State on failover	"Standby-region state is permanently lost"	Fully preserved

Microsoft documents that recovery on Azure is reconstruction, not failover: "Warm standby environments start mostly empty. Recovery is reconstruction, not promotion of a hot replica," and recovered agents "have no access to prior threads," with "no state… transferable between regions." (Platform outage recovery)

Azure platform outages are a documented reality

A regional event is not a tail risk for an agent with no automatic failover and a "total loss" recovery point. **Oct 29, 2025** — an Azure Front Door global configuration fault broke routing and authentication across Azure-fronted services for hours. **May 29, 2026** — Azure OpenAI Service had a multi-region outage (failures, timeouts, 5XX) with pronounced impact across Europe and Australia East, from retry amplification cascading across regions. (Azure status history)

Cost: Many Meters vs. One Fee

AI systems built on Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required to run the same agentic transaction volume, not list price. On Foundry, cost is the sum of independent meters: per-token or PTU-reserved OpenAI, per-RU Cosmos DB (≥3,000 RU/s minimum), per-tier AI Search, gateway units, and per-GB log ingestion.

PTU reservations carry a substantial monthly floor; independent 2026 pricing analyses put a single minimum PTU reservation in the low thousands of dollars per month before the surrounding services are counted (CloudZero). Each meter scales independently, making the total bill hard to forecast.

Akka runs orchestration, agents, memory, streaming, APIs, observability, and governance on **shared compute** under a **fixed annual fee**. The efficiency is structural: actor-based concurrency delivers ~10 trillion tokens per core per year versus ~2 trillion for comparable solutions, and ~80% less compute than Python-based frameworks. Manulife reported up to **300% more concurrency** and **30–50% faster processing** after porting Python-based systems to Akka. (akka.io/blog/go-slow-to-go-fast)

Continuous AI Intelligence

Akka Optimize keeps improving the AI after it ships. It runs an evaluation loop on your own production traffic, grading live interactions and training smaller specialized models on your proprietary data, so the system keeps getting better as it runs. The loop delivers three results that compound over time:

- **Model leverage** — route each request to the best model from any vendor under your policies, and reserve frontier models for the requests that need them.
- **Data sovereignty** — the specialized models are trained on your proprietary data, owned by you, and run inside your own environment.
- **Cost governance** — full visibility and control of AI spend, with savings that compound as the loop keeps running.

The evaluations run continuously inside Akka and grade traffic from agents wherever they run, in Akka or in third-party harnesses.

Governance and the EU AI Act: Assessment vs. Inline Enforcement

The penalties are enforceable now

Violation	Maximum Fine
Prohibited AI practices (Art. 5)	EUR 35M or 7% global turnover
High-risk obligations (Art. 9-15)	EUR 15M or 3% global turnover
Supplying incorrect information	EUR 7.5M or 1.5% global turnover

Prohibited-practice penalties enforceable since Feb 2025, high-risk since Aug 2025; high-risk AI carries a 10-year logging-retention obligation (Art. 72).

Microsoft has real assessment tooling: Purview Compliance Manager ingests the EU AI Act, extracts controls with AI, maps improvement actions, and syncs Foundry evaluation results. (Purview + Foundry) Foundry guardrails check an agent's input, tool calls, tool responses, and output and block content that breaks a control, for agents built in Foundry Agent Service only; guardrails for agents are in preview. Control Plane lets an administrator stop a hosted agent or block requests to a registered agent.

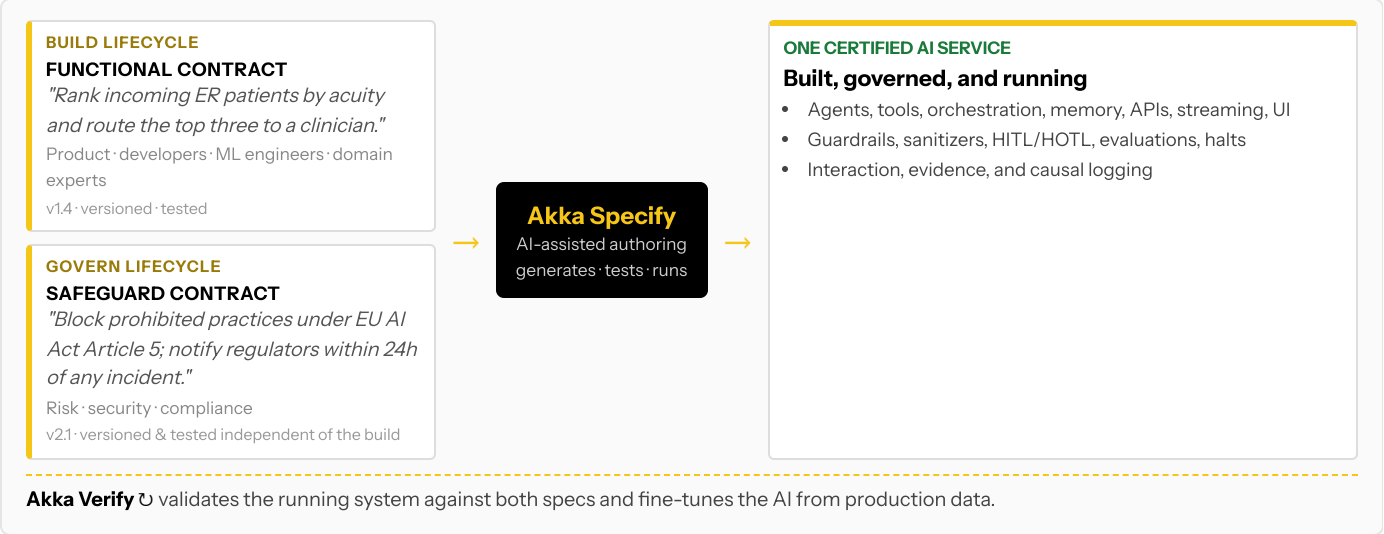
Requirement	Azure approach — the gap
Real-time policy enforcement	Foundry guardrails at input, tool call, tool response, and output (preview); Foundry Agent Service agents only
Decision explainability	Post-hoc evaluations, not an inline record of a running decision
Immutable interaction log	Standard Monitor/Purview logs, not a hash-chained ledger
Human pause / override	Control Plane stops hosted and published agents and blocks custom agents; a stop does not end runs in progress
Authorization capture	Correlate Entra + RBAC + gateway logs, not captured atomically
PII scrub with explain	Detection exists; no single atomic scrub-and-explain
Pre-deployment classification + sign-off	Compliance Manager templates and AI Red Teaming Agent scans, not a runtime-gating attestation engine

Sealed audit artifact	Assembled from reports; no per-deployment sealed package

Akka governance is **aspect-woven into the runtime**: inline guardrails, policies, LLMs-as-a-judge, and sanitizers execute within the agents; evidence is hash-chained and immutable; humans can pause, override, or nudge a running process; PII is scrubbed and explained atomically. Before a system ships, Akka classifies it against **241 regulations and frameworks and 1,718 controls (700 carrying financial penalties)**, routes change events to the right reviewers, and emits a sealed Governance Posture Package. Akka Verify proves conformance from the running system.

Two Lifecycles, One Certified System

Building on Foundry means developers assembling agents against Azure SDKs and portals. Administrators can define versioned guardrail policies in Foundry Control Plane, and those guardrails apply to agents built in Foundry Agent Service. Akka runs two independent lifecycles on one platform via **Akka Specify**:



The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences. Foundry has no equivalent governance authoring lifecycle that produces a certified, sealed system.

Real-Time Streaming at Petabyte Scale

Foundry has no built-in streaming engine; real-time pipelines are provisioned separately and operated by the customer. Akka streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

For the Buyer: A Recent, Renamed Product vs. an 18-Year Platform

Microsoft is a durable company; the question is the maturity of the **product**. The current Foundry Agent Service reached general availability in March 2026, after an earlier release at Build in May 2025, and the product is mid-rebrand from "Azure AI Foundry" to "Microsoft Foundry," with RBAC roles renamed from "Azure AI User/Owner/Project Manager" to "Foundry User/Owner/Project Manager" as the change rolls out. (HA/DR for Foundry) It is a young, evolving agent layer assembled over a set of Azure services, and Foundry provides no automatic failover or disaster recovery for it.

Buyer concern	Foundry	Akka
Product maturity	GA March 2026 (first release May 2025); active rebrand (Azure AI Foundry → Microsoft Foundry) and RBAC rename in progress	since 2009; 100,000+ production deployments; 52 banks; 2B+ people reached daily
Scope of accountability	Customer integrates and operates Cosmos DB, AI Search, Storage, gateway, OpenAI; customer owns durability of stateful dependencies	One platform, one SLA, 24/7 SRE — Akka owns the running system
Risk transfer	Standard Azure terms	Availability and data-integrity guarantees backed by contractual indemnities
HA / DR	No automatic failover; "total loss" recovery point	Active-active; sub-1-min RTO; zero-byte RPO

Certifications & audits	Inherited Azure certifications + Purview Compliance Manager templates	19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io)
Budget predictability	Several independent meters that scale with load	Fixed annual fee on shared compute

The decision is scope and accountability: Foundry gives you an agent service to wire your Azure estate around; Akka gives you the integrated platform with the SLA on the whole thing.

Customers Running Agentic and Real-Time Systems on Akka

Manulife
Agentic AI
selected Akka (Mar 2026) to operationalize agentic AI with regulated-grade governance

Tubi
5B tok/s
real-time hyper-personalization engine

Swiggy
71ms
order-assignment AI, ~50% latency reduction

John Deere
1,000+
tractor sensors turned into real-time insight

Verizon
750%
order-processing capacity gain; 6s → 2.4s response

Common Questions

We're already on Azure. Why add Akka?

Akka deploys inside your Azure VPC. You keep your Azure infrastructure, your Entra ID, and your networking — Akka runs alongside them and adds the integrated runtime, active-active HA/DR, the 99.9999% SLA, and governance for AI systems beyond those built in Foundry Agent Service.

Won't Microsoft close the HA/DR gap soon?

Microsoft's current documentation (updated May–September 2026) states that Foundry has no automatic failover, that the Agent Service has no built-in DR, and that the recovery point "can be total loss." Until that changes in the product, the availability of an assembled Foundry agent is the customer's responsibility. Akka delivers active-active HA/DR with a contractual SLA today.

Doesn't Microsoft already cover the EU AI Act?

Microsoft has assessment tooling and inline guardrails. Purview Compliance Manager has templates for the EU AI Act, ISO/IEC 42001, ISO/IEC 23894, and NIST AI RMF and syncs Foundry evaluation results. Foundry guardrails block content at agent tool calls (preview), for agents built in Foundry Agent Service only. The Act expects immutable records witnessed as they happen, human override of a running process, authorization capture at execution time, and pre-deployment classification that gates a release. Akka enforces these in the runtime and produces a sealed posture package.

Isn't Foundry cheaper because we have an EA?

EA discounts apply to individual Azure services, but a production agent still meters across Azure OpenAI, Cosmos DB, AI Search, Storage, the gateway, and logging — each scaling independently. The Akka shared-compute model runs all of it on one runtime at a fixed annual fee, and is up to 90% cheaper to operate for the same agentic transaction volume.

Sources

Foundry HA/DR (no failover): learn.microsoft.com/en-us/azure/foundry/how-to/high-availability-resiliency (updated 2026-09-21) — "Foundry itself doesn't provide automatic failover or disaster recovery"; customer owns durability of Cosmos DB / AI Search / Storage

Foundry Agent Service DR: learn.microsoft.com/en-us/azure/foundry/how-to/agent-service-disaster-recovery (updated 2026-05-12) — "doesn't replicate state, create backups, or support point-in-time restore"; "no supported method for active-active, multi-region replication"; "recovery point for stateful content can be total loss"

Foundry platform outage recovery: learn.microsoft.com/en-us/azure/foundry/how-to/agent-service-platform-disaster-recovery — RTO "30 or more minutes per project"; "reconstruction, not promotion of a hot replica"; "standby-region state is permanently lost"

Standard agent setup (required services): learn.microsoft.com/en-us/azure/foundry/agents/concepts/standard-agent-setup — Cosmos DB (≥3,000 RU/s), AI Search, Storage, Key Vault required; plus OpenAI, API Management, Monitor

Foundry Agent Service GA: techcommunity.microsoft.com — GA announced at Build, May 2025; devblogs.microsoft.com/foundry/foundry-agent-service-ga — GA of the current Agent Service, March 16, 2026

Purview + Foundry governance: learn.microsoft.com/en-us/purview/ai-azure-foundry — Compliance Manager extracts EU AI Act controls, maps improvement actions, syncs Foundry evaluations (assessment/mapping); learn.microsoft.com/en-us/azure/foundry/guardrails/guardrails-overview — agent guardrails (preview); learn.microsoft.com/en-us/azure/foundry/control-plane/how-to-manage-agents — start, stop, block agents; learn.microsoft.com/en-us/purview/compliance-manager-assessments — AI regulation templates

Azure OpenAI / PTU pricing (third-party): cloudzero.com/blog/azure-openai-pricing; opslyft.com/blog/azure-openai-pricing — PTU minimum reservations and multi-meter cost

Azure outages: azure.status.microsoft/en-us/status/history — Oct 29 2025 Front Door global fault; May 29 2026 Azure OpenAI multi-region outage (Europe / Australia East)

Akka performance: akka.io/blog/go-slow-to-go-fast — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

Akka platform: 99.9999% availability, active-active HA/DR, sub-1-min RTO, zero-byte RPO (contractual indemnities); 241 regulations and frameworks / 1,718 controls / 699 with penalties; 100,000+ deployments / since 2009; profitable; Dell Technologies Capital

Akka trust center: trust.akka.io — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

Manulife: akka.io/blog/manulife-selects-akka-to-operationalize-agentic-ai (March 10, 2026)